

Hybrid Time Formulation for Diesel Blending and Distribution Scheduling

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ABSTRACT: This work focuses on the scheduling of an in-line diesel blending and distribution subsystem of an oil refinery. The formulation is based on a hybrid time representation in which time points are equally distributed along the time horizon, within which time slots of variable length are postulated. The hybrid time representation takes advantage of the flexibility of the continuous time representation and enables handling of intermediate due dates with the use of fixed time points. Time variables are defined in terms of resources instead of transfer operations leading to a smaller size formulation. Results from a real-world case are used to validate the proposed formulation that takes into consideration capacities and operating rules while minimizing costs. Results of the multiperiod scheduling are also compared to a day-ahead production planning policy. Computational efficiency is achieved after adding valid inequalities and symmetry breaking constraints.

1. INTRODUCTION

Undoubtedly, enterprisewide optimization is the most efficient way to leverage companies to their highest degree of performance. However, there remains a significant technological and organizational gap between where we are today and where we want to be in the future. Besides, merging subsystems into a single system requires understanding beyond each individual subsystem. Particularly, scheduling in oil refineries is too complex to be optimized as a monolithic model; hence it is normally decomposed at the subsystem level. Typically the three subsystems addressed independently in scheduling problems include the following: (1) crude oil unloading, mixing, and inventory control; (2) production unit scheduling; and (3) finished product blending and shipping. There have been efforts that address more than one refinery subsystem simultaneously.^{1–3} However, simplifying assumptions are still necessary to be able to deal with real-world problems. In the last three decades, a lot of attention has been devoted in studying the first subsystem.^{4–12} The reported literature has comprised subjects related to problem features, time representation, and formulation of tighter models and solution strategies. Notably, significantly less attention has been devoted to the problems pertaining to the third subsystem.^{13–16} Generally, optimizing operations related to the end products of the refinery result in benefits that include the following:

- efficient use of the intermediate products;
- decreased product giveaway;
- elimination of off-spec blends;
- inventory control in the light of upcoming demands and disturbances to the system;
- minimization of losses resulting from distribution.

In this work, we address the diesel blending and distribution scheduling problem. Its major challenge involves in-line blending, which requires simultaneous alignment of multiple tanks to the same pipeline so that a product is blended and shipped without a storage step. Simultaneous alignment of tanks to a pipeline for the duration of the blending operation guarantees homogeneity of the end product composition. Modeling such a problem with a

discrete time representation is quite straightforward but the problem rapidly becomes intractable as the time horizon is expanded. The continuous time representation is an evident candidate, but its drawback relies on the difficulty of defining intermediate due dates.¹⁷ The hybrid time representation then comes into place. Discrete fixed points circumscribe by when events should occur like demand satisfaction, whereas variable slots in between the fixed points are used to create operations whose length are determined by the optimization problem.^{18,19} Some researchers have explored in detail time representation and solution efficiency of scheduling problems for both batch and continuous processes,¹¹ whereas some focus on particular industries such as oil refineries.¹² Two recent comprehensive reviews^{20,21} present a detailed discussion of time representation for production scheduling models in process systems engineering.

In this paper we propose a novel formulation that is based on the proposal of Hu and Zhu (as cited in the 2012 work of Chen et al.¹²), in which timing decisions are resource-centric instead of operation-centric, thus rendering smaller size models. This approach complies with the requirement that overlapping operations should be assigned to the same event point and yet not necessarily to have coincident time lengths considering the different resources involved, as is the in-line blending operation. The model also relies on a hybrid time representation composed of event points that are grouped in disjunctive sets that are assigned to each day. Moreover, event points are defined in monotonically increasing order so that transitions in the pipeline can be identified by evaluating only adjacent event points, regardless of which days they belong to. Moreover, with this approach, transition identification between different days does not require additional variables.

Special Issue: Jaime Cerdá Festschrift

Received: March 8, 2014

Revised: June 12, 2014

Accepted: June 13, 2014

Published: June 13, 2014



The text is organized as follows: in section 2, the problem description is provided. In section 3 the adopted time representation is detailed, whereas the proposed formulation is presented in section 4. Results and discussions are presented in section 5. Finally, concluding remarks are brought up in section 6.

2. PROBLEM DESCRIPTION

The first step in oil processing is its fractionation into a series of intermediate streams. The less noble intermediate streams are then further processed in downstream units with the intent to break the long chain molecules into more valuable products or undergo further purification. Finished products are obtained by mixing selected intermediate streams. Product qualities are strongly dependent on the blending of the intermediate streams, whose qualities in turn depend on crude oil properties, operating modes, and feed conditions of the involved processing units. Rigorous mathematical representation of process units results in nonlinear models which commonly incorporate bilinear terms, leading to nonconvex problems. Therefore, in order to avoid excessive complex models, simplifications are commonly made.

Figure 1 depicts the general structure of the blending and distribution problem addressed by the present work. A set of

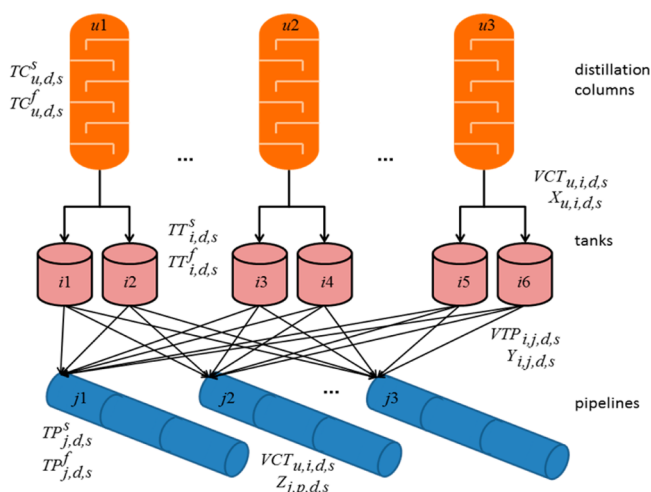


Figure 1. Schematic of diesel in-line blending and distribution infrastructure.

distillation units produce intermediate streams that are sent to dedicated storage tanks. It is assumed that qualities of the streams generated by the distillation units remain constant throughout the time horizon. Given that tanks are not allowed to load and unload simultaneously and that distillation units operate continuously, each distillation unit discharges to two rundown tanks, albeit not simultaneously. However, at any time there is only a one-to-one alignment between a distillation unit and a tank. Therefore, a tank might be either loading, unloading or idle. The content of each pair of tanks is undistinguishable and is the feedstock for producing diesel with three different grades. Each diesel grade is differentiated by two quality indicators: sulfur content and cetane number. Blending is accomplished in-line just prior to feeding pipelines; in other words, there is no storage facility between the blending point and pipelines. Pipelines operate independently and under unique demand requirements. Diesel parcels with different grades are

pumped contiguously, in which case an interface between two adjacent parcels establish an undesirable interface containing off-spec material demanding appropriate handling and incurring costs.

Given the foregoing description the optimization problem is concerned with determining the following:

- Tank operations management;
- Tank inventory profiles;
- Intermediate product consumption and blending scheduling;
- End product shipment scheduling.

3. HYBRID TIME REPRESENTATION

In this work, time is represented through a hybrid, discrete and continuous representation. Time horizon is partitioned into periods of equal length representing days at which points demand may be incurred. Within each day a prespecified number of time slots is postulated, whose duration is to be determined by the optimization problem. The set of slots is subdivided in mutually exclusive subsets and assigned to each of the days. Consequently, slots used to represent events in the first day are not the same as those used in the second day and so forth. Moreover, throughout the time horizon slots are placed in monotonically increasing order. Figure 2a illustrates the general framework. Resource-centric representations are considered for the continuous timing decisions, Figure 2b. Overlapping operations must be assigned to the same time slot in each day, however, neither durations nor start/finish times need to coincide. Further details are presented in the modeling section concerning timing constraints.

4. MATHEMATICAL FORMULATION

The proposed formulation relies on the terms listed in the Nomenclature.

Material Balances and Capacity Constraints. Equation 1 represents the mass balance for each tank at the end of each day. The volume contained in tank i at the end of day d is given in terms of the volume contained in that tank at the end of the previous day plus the amounts of materials transferred from distillation units to the corresponding tank, minus the amounts of materials transferred to pipelines.

$$V_i^{\min} \leq VD_{i,d} = VD_{i,d-1} + \sum_{u \in U_i} \sum_{s \in SD_d} VCT_{u,i,d,s} - \sum_{j \in J_i} \sum_{s \in SD_d} VTP_{j,i,d,s} \leq V_i^{\max} \quad \forall i, d \quad (1)$$

For $d = 1$, variable $VD_{i,d-1}$ in constraint 1 is replaced by the initial inventory that is denoted by $VD_{i,0}$.

For all slots but the first, eq 2 calculates the volume contained in tank i at the end of time slot s of day d in terms of the volume contained in that tank at the end of the previous slot plus the amounts of materials transferred from distillation units to the corresponding tank, minus the amounts of materials transferred to pipelines.

$$V_i^{\min} \leq VS_{i,d,s} = VS_{i,d,s-1} + \sum_{u \in U_i} \sum_{\substack{s' \in SD_d \\ s' \leq s}} VCT_{u,i,d,s'} - \sum_{j \in J_i} \sum_{\substack{s' \in SD_d \\ s' \leq s}} VTP_{j,i,d,s'} \leq V_i^{\max} \quad \forall i, d, s \in SD_d - FS_d \quad (2)$$

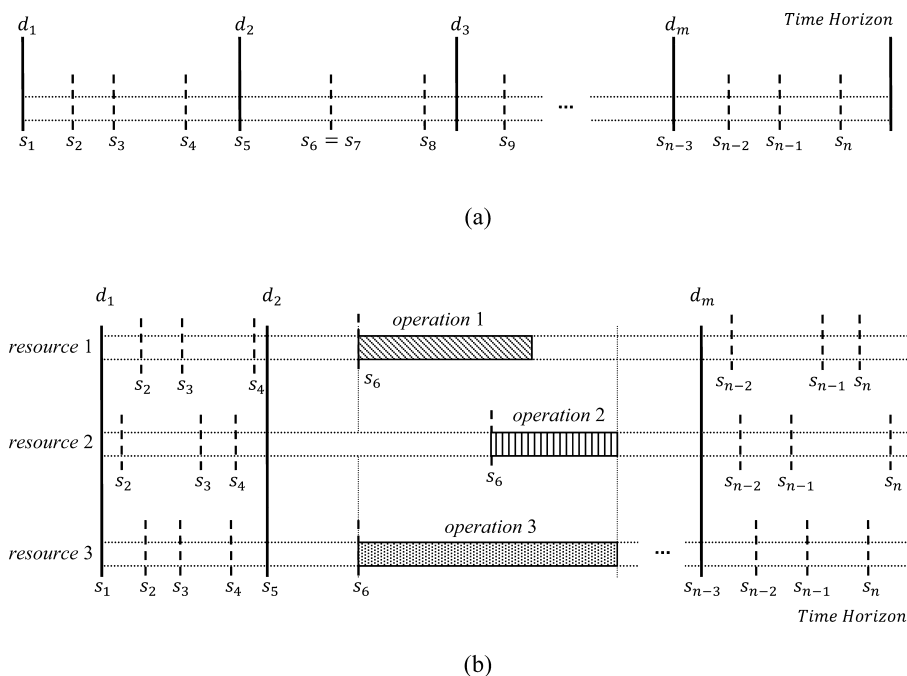


Figure 2. Time representation: (a) hybrid time representation framework and (b) resource-centric representation for continuous timing decisions.

Tracking the inventory at the end of each day is necessary in order to be able to account for the inventory at the transition from 1 day to another, as given by eq 3.

$$V_i^{\min} \leq VS_{i,d,s} = VD_{i,d-1} + \sum_{u \in \mathbf{U}_i} \sum_{\substack{s' \in \mathbf{SD}_d \\ s' \leq s}} VCT_{u,i,d,s'} - \sum_{j \in \mathbf{J}_i} \sum_{\substack{s' \in \mathbf{SD}_d \\ s' \leq s}} VTP_{j,i,d,s'} \leq V_i^{\max} \quad \forall i, d, s \in \mathbf{FS}_d \quad (3)$$

Similarly to constraint 1, $VD_{i,d-1}$ in constraint 3 is replaced by $VD_{i,0}$ when $d = 1$.

The volumetric balance between tanks and pipelines is established through eq 4. The sum of all parcels withdrawn from tanks (right-hand side of the equation) must equal the amount of end products blended and shipped through pipeline j . As will be discussed later, only one end product is blended at a time, which results in having only one nonzero term on the right-hand side of 4.

$$\sum_{i \in \mathbf{I}_j} VTP_{i,j,d,s} = \sum_{p \in \mathbf{P}} VPP_{j,p,d,s} \quad \forall j, d, s \in \mathbf{SD}_d \quad (4)$$

Constraint 5 imposes restrictions on product specification assuming that all product properties obey a volumetric linear mixing rule. Note that 5 can be written without loss of generality as a less than or equal to constraint by replacing $SPC_{p,k}$ by $(-SPC_{p,k})$.

$$\sum_{i \in \mathbf{I}_j} K_{i,k} VTP_{i,j,d,s} \geq \sum_{p \in \mathbf{P}} SPC_{p,k} VPP_{j,p,d,s} \quad \forall j, k, d, s \in \mathbf{SD}_d \quad (5)$$

The accumulated amount of each end product dispatched through each pipeline must completely fulfill the accumulated daily demand, as given by constraint 6. However, along the whole time horizon the accumulated amount of each product must be exactly fulfilled (eq 7).

$$\sum_{\substack{d' \in \mathbf{D} \\ d' \leq d}} \sum_{s' \in \mathbf{SD}_{d'}} VPP_{j,p,d',s'} \geq \sum_{\substack{d' \in \mathbf{D} \\ d' \leq d}} D_{j,p,d'} \quad \forall j, p, d \in \mathbf{D} \quad (6)$$

$$\sum_{d \in \mathbf{D}} \sum_{s \in \mathbf{SD}_d} VPP_{j,p,d,s} = \sum_{d \in \mathbf{D}} D_{j,p,d} \quad \forall j, p \quad (7)$$

Upper and lower bounds on the transferred volumes between each pair of resources are established through constraints 8–10, which are only enforced once the corresponding binary variables are active. Otherwise, these constraints are relaxed and in which case they do not even impose that the transferred volumes should be null. Therefore, constraints 11–13 are necessary to complement constraints 8–10.

$$FC_u^{\min}(TC_{u,d,s}^f - TC_{u,d,s}^s) - VC_u^{\text{up}}(1 - X_{u,i,d,s}) \leq VCT_{u,i,d,s} \leq FC_u^{\max}(TC_{u,d,s}^f - TC_{u,d,s}^s) \quad \forall u, i \in \mathbf{I}_w, d, s \in \mathbf{SD}_d \quad (8)$$

$$FT_i^{\min}(TT_{i,d,s}^f - TT_{i,d,s}^s) - VT_i^{\text{up}}(1 - Y_{i,j,d,s}) \leq VTP_{i,j,d,s} \leq FT_i^{\max}(TT_{i,d,s}^f - TT_{i,d,s}^s) \quad \forall i, j \in \mathbf{J}_i, d, s \in \mathbf{SD}_d \quad (9)$$

$$FP_j^{\min}(TP_{j,d,s}^f - TP_{j,d,s}^s) - VP_j^{\text{up}}(1 - Z_{j,p,d,s}) \leq VPP_{j,p,d,s} \leq FP_j^{\max}(TP_{j,d,s}^f - TP_{j,d,s}^s) \quad \forall j, p, d, s \in \mathbf{SD}_d \quad (10)$$

$$VCT_{u,i,d,s} \leq HDFT_u^{\max} X_{u,i,d,s} \quad \forall u, i \in \mathbf{I}_w, d, s \in \mathbf{SD}_d \quad (11)$$

$$VTP_{i,j,d,s} \leq HDFT_i^{\max} Y_{i,j,d,s} \quad \forall i, j \in \mathbf{J}_i, d, s \in \mathbf{SD}_d \quad (12)$$

$$VPP_{j,p,d,s} \leq HDFT_j^{\max} Z_{j,p,d,s} \quad \forall j, p, d, s \in \mathbf{SD}_d \quad (13)$$

Note that timing variables are associated with resources instead of transfer operations that link source and sink points,

as proposed by Hu and Zhu (as cited in the 2012 work of Chen et al.¹²). For instance, if one takes a tank, the same start and end times related to that tank may be either allocated to a loading or unloading operation. Moreover, if a tank unloads, there is no explicit reference to which pipeline that tank feeds to by inspecting the timing variables. The benefit of not indexing the timing variables to transfer operations relies on the smaller number of variables and time slots required to accommodate a problem, normally resulting in models with smaller dimension.¹²

Operating Rules. Distillation units discharge to at most one rundown tank as imposed by 14. Moreover, simultaneous loading and unloading is forbidden. Therefore, if a tank is charged, 15 sets $\sum_{j \in J_i} Y_{ij,d,s} = 0$. If, on the other hand, a tank is unloading, then the maximum number of pipelines aligned to that tank is NC_i .

$$\sum_{i \in I_u} X_{u,i,d,s} \leq 1 \quad \forall u, d, s \in \mathbf{SD}_d \quad (14)$$

$$\sum_{j \in J_i} Y_{ij,d,s} \leq NC_i(1 - X_{u,i,d,s}) \quad \forall u, i \in I_u, d, s \in \mathbf{SD}_d \quad (15)$$

Given that blending is accomplished in-line prior to feeding a pipeline, at most one product can be blended at a time and shipped through the pipeline. Therefore, 16 assures that this operational constraint is always satisfied.

$$\sum_{p \in P} Z_{j,p,d,s} \leq 1 \quad \forall j, d, s \in \mathbf{SD}_d \quad (16)$$

Also, an end product cannot be shipped from a given pipeline more than once within a day, as given by constraint 17.

$$\sum_{s \in \mathbf{SD}_d} Z_{j,p,d,s} \leq 1 \quad \forall j, p, d \quad (17)$$

Timing Constraints. It is important to note that from constraints 8–13 even if no operation is assigned to a resource a non null time window might still be allocated to that resource. Constraints 18–20 are added to avoid such situation.

$$(TC_{u,d,s}^f - TC_{u,d,s}^s) \leq HD \sum_{i \in I_u} X_{u,i,d,s} \quad \forall u, d, s \in \mathbf{SD}_d \quad (18)$$

$$(TT_{i,d,s}^f - TT_{i,d,s}^s) \leq HD(X_{u,i,d,s} + \sum_{j \in J_i} Y_{ij,d,s}) \quad \forall i, u \in U_i, d, s \in \mathbf{SD}_d \quad (19)$$

$$(TP_{j,d,s}^f - TP_{j,d,s}^s) \leq HD \sum_{p \in P} Z_{j,p,d,s} \quad \forall j, d, s \in \mathbf{SD}_d \quad (20)$$

Note that if there are no operations assigned between a column and its rundown tanks ($\sum_{j \in J_i} X_{ij,d,s} = 0$) for a given time slot, constraint 18 enforces the column operating window to be null in that slot. Likewise, if tanks are idle ($X_{u,i,d,s} + \sum_{j \in J_i} Y_{ij,d,s} = 0$) or if pipelines are idle ($\sum_{p \in P} Z_{j,p,d,s} = 0$), constraints 19 and 20 enforce tank and pipeline time windows to be null, respectively.

In addition to the upper bound on the time window assigned to an operation, a lower bound is also considered in order to avoid very short operations that are not feasible from a practical

perspective (constraints 21–23). Constraint 22 does not have to be defined in terms of the loading operations, i.e., 22 does not need to incorporate a term establishing a relationship with the $X_{u,i,d,s}$ variables due to the fact that distillation units feed a single tank at a time and that their time windows are synchronized. Therefore, as will be discussed in the Timing Constraints section, when a tank loads, its time window exactly matches the time window of the distillation unit discharging to that tank.

$$(TC_{u,d,s}^f - TC_{u,d,s}^s) \geq MH \sum_{i \in I_u} X_{u,i,d,s} \quad \forall u, d, s \in \mathbf{SD}_d \quad (21)$$

$$(TT_{i,d,s}^f - TT_{i,d,s}^s) \geq MH Y_{ij,d,s} \quad \forall i, j \in J_i, d, s \in \mathbf{SD}_d \quad (22)$$

$$(TP_{j,d,s}^f - TP_{j,d,s}^s) \geq MH \sum_{p \in P} Z_{j,p,d,s} \quad \forall j, d, s \in \mathbf{SD}_d \quad (23)$$

Since distillation units must operate continuously, constraint 24 is required to enforce operation continuity.

$$\sum_{s \in \mathbf{SD}_d} (TC_{u,d,s}^f - TC_{u,d,s}^s) = HD \quad \forall u, d \quad (24)$$

Timing variables must be monotonically increasing (constraints 25–30) and must be defined within the domain of each day, i.e., between the first and the last hours of each day (constraints 31–36).

$$TC_{u,d,s+1}^s \geq TC_{u,d,s}^f \quad \forall u, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (25)$$

$$TC_{u,d,s}^f \geq TC_{u,d,s}^s \quad \forall u, d, s \in \mathbf{SD}_d \quad (26)$$

$$TT_{i,d,s+1}^s \geq TT_{i,d,s}^f \quad \forall i, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (27)$$

$$TT_{i,d,s}^f \geq TT_{i,d,s}^s \quad \forall i, d, s \in \mathbf{SD}_d \quad (28)$$

$$TP_{j,d,s+1}^s \geq TP_{j,d,s}^f \quad \forall j, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (29)$$

$$TP_{j,d,s+1}^f \geq TP_{j,d,s}^s \quad \forall j, d, s \in \mathbf{SD}_d \quad (30)$$

$$T_d^f \leq TC_{u,d,s}^s \leq T_d^l \quad \forall u, d, s \in \mathbf{SD}_d \quad (31)$$

$$T_d^f \leq TC_{u,d,s}^f \leq T_d^l \quad \forall u, d, s \in \mathbf{SD}_d \quad (32)$$

$$T_d^f \leq TT_{i,d,s}^s \leq T_d^l \quad \forall i, d, s \in \mathbf{SD}_d \quad (33)$$

$$T_d^f \leq TT_{i,d,s}^f \leq T_d^l \quad \forall i, d, s \in \mathbf{SD}_d \quad (34)$$

$$T_d^f \leq TP_{j,d,s}^s \leq T_d^l \quad \forall j, d, s \in \mathbf{SD}_d \quad (35)$$

$$T_d^f \leq TP_{j,d,s}^f \leq T_d^l \quad \forall j, d, s \in \mathbf{SD}_d \quad (36)$$

Constraints 37 and 38 synchronize start times between columns and tanks, whereas 39 and 40 synchronize their end times. If a column discharges to two different tanks sequentially, each transfer is assigned to a different time slot (see Figure 3).

$$TC_{u,d,s}^s \geq TT_{i,d,s}^s - H_d^f(1 - X_{u,i,d,s}) \quad \forall u, i \in I_u, d, s \in \mathbf{SD}_d \quad (37)$$

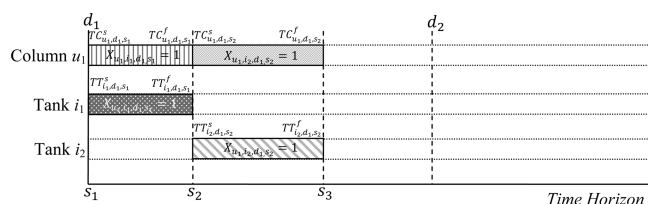


Figure 3. Synchronization of time windows of resources that are not allowed to overlap.

$$TC_{u,d,s}^s \leq TT_{i,d,s}^s + H_d^f(1 - X_{u,i,d,s})$$

$$\forall u, i \in I_u, d, s \in SD_d \quad (38)$$

$$TC_{u,d,s}^f \geq TT_{i,d,s}^f - H_d^f(1 - X_{u,i,d,s})$$

$$\forall u, i \in I_u, d, s \in SD_d \quad (39)$$

$$TC_{u,d,s}^f \leq TT_{i,d,s}^f + H_d^f(1 - X_{u,i,d,s})$$

$$\forall u, i \in I_u, d, s \in SD_d \quad (40)$$

As for transfers between tanks and pipelines, given that a tank can feed multiple pipelines at the same time, the start and end times of the involved resources do not have to coincide. The only requirement is that if multiple tanks are aligned to a pipeline within the same time slot, the time slot of the pipeline must be completely contained by the time slots of each tank. Constraints 41 and 42 guarantee that all required tanks feed the pipeline from the beginning to the end of the blending operation, which guarantees product homogeneity.

$$TT_{i,d,s}^s \leq TP_{j,d,s}^s - H_d^f(1 - Y_{i,j,d,s})$$

$$\forall i, j \in J_i, d, s \in SD_d \quad (41)$$

$$TT_{i,d,s}^f \geq TP_{j,d,s}^f - H_d^f(1 - Y_{i,j,d,s})$$

$$\forall i, j \in J_i, d, s \in SD_d \quad (42)$$

Blending is executed by aligning multiple tanks to the same pipeline simultaneously. Inequalities 41 and 42, however, do not enforce that tank time windows have exactly the same length or equal start and end times of those of the pipelines, as seen by the example illustrated in Figure 4 (note that Tank 1 is

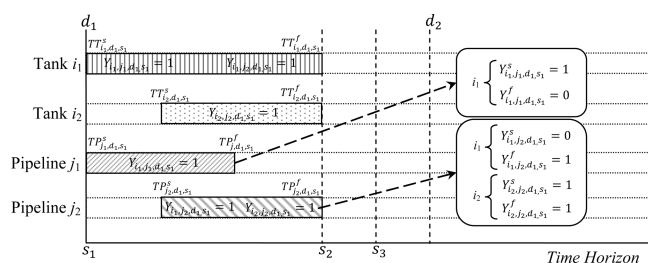


Figure 4. Time window synchronization of resources that are allowed to overlap.

assigned to the two pipelines but its start time does not match that of Pipeline 2).

In order to guarantee that the time window related to a tank unloading operation exactly matches time windows of multiple pipelines, binary variables $Y_{i,j,d,s}^s$ and $Y_{i,j,d,s}^f$ are incorporated to define which pipeline operation whose start time should exactly

match the start time of the tank unloading operation and which pipeline operation whose end time should exactly match the end time of the tank unloading operation, respectively. Moreover, constraints 43 and 44 are added to enforce that time slots of tanks related to unloading operations should not extend beyond time slots of pipelines, whereas 45 imposes that if a tank feeds a pipeline, there must be at least one pipeline operation whose start time is coincident with the start time of a tank unloading operation. Likewise, 46 applies for the end time operations. Constraints 47 and 48 are added to certify that variables $Y_{i,j,d,s}^s$ and $Y_{i,j,d,s}^f$ will only be active if a tank unloading operation is active.

$$TT_{i,d,s}^s \geq TP_{j,d,s}^s - H_d^f(1 - Y_{i,j,d,s}^s)$$

$$\forall i, j \in J_i, d, s \in SD_d \quad (43)$$

$$TT_{i,d,s}^f \leq TP_{j,d,s}^f - H_d^f(1 - Y_{i,j,d,s}^f)$$

$$\forall i, j \in J_i, d, s \in SD_d \quad (44)$$

$$\sum_{j' \in J_i} Y_{i,j',d,s}^s \geq Y_{i,j,d,s}^s \quad \forall i, j \in J_i, d, s \in SD_d \quad (45)$$

$$\sum_{j' \in J_i} Y_{i,j',d,s}^f \geq Y_{i,j,d,s}^f \quad \forall i, j \in J_i, d, s \in SD_d \quad (46)$$

$$Y_{i,j,d,s}^s \leq Y_{i,j,d,s} \quad \forall i, j \in J_i, d, s \in SD_d \quad (47)$$

$$Y_{i,j,d,s}^f \leq Y_{i,j,d,s} \quad \forall i, j \in J_i, d, s \in SD_d \quad (48)$$

Transition Identification. When an end product is pumped in a pipeline adjacently to another product, an interface composed of their mixture is formed. Since there is always degradation involved with the mixture of end products, such transitions are certainly undesirable.

Transitions may be identified by inspecting consecutive time slots as in constraint 49. The constraint states that if end product p' is shipped after end product p in adjacent time slots, a transition from p' to p must be identified ($TR_{j,p',p,s+1} = 1$) in that pipeline.

$$TR_{j,p',p,s+1} \geq Z_{j,p',d',s+1} + Z_{j,p,d,s} - 1$$

$$\forall j, p, p', p \neq p', d' \in DS_{s+1}, d \in DS_s, s < |S| \quad (49)$$

However, transition identification fails in case no product is allocated to either of the slots under consideration. Moreover, transitions incur cost so that the optimization will always favor empty slots. Therefore, a different mechanism of identifying transitions must be devised.

In order to circumvent this issue, a new set of binary variables $W_{j,p,d,s}$ is defined, whose value is nonzero when no product is allocated to time slot s and the last shipped product is p . Constraint 50 relates $Z_{j,p,d,s}$ and $W_{j,p,d,s+1}$ and also guarantees that the logic is enforced when two or more consecutive time slots receive no product allocation—in this case, $W_{j,p,d,s+1} \geq W_{j,p,d,s}$. Constraint 51 enforces that either $Z_{j,p,d,s}$ or $W_{j,p,d,s}$ is nonzero in each time slot. Finally, constraint 49 is modified into constraint 52, through which transitions are identified. Given that time slots are distributed sequentially throughout the time horizon, constraint 52 accounts for intra- and interday transitions.

$$\sum_{p'} Z_{j,p',d,s+1} + W_{j,p,d,s+1} \geq W_{j,p,d,s} + Z_{j,p,d,s} \quad \forall j, p, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (50)$$

$$\sum_p Z_{j,p,d,s} + \sum_p W_{j,p,d,s} = 1 \quad \forall j, d, s \in \mathbf{SD}_d \quad (51)$$

$$TR_{j,p,p',s+1} \geq Z_{j,p',d',s+1} + (W_{j,p,d,s} + Z_{j,p,d,s}) - 1 \quad \forall j, p, p', p' \neq p, d' \in \mathbf{DS}_{s+1}, d \in \mathbf{DS}_s, s < |\mathbf{S}| \quad (52)$$

Consider the example from Figure 5. Product p_2 is allocated to time slot s_2 in the first day and there is no other allocation

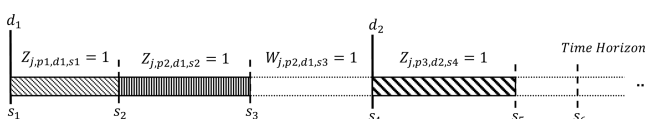


Figure 5. Example of transition identification.

within that day after dispatch of p_2 has been completed. Therefore, $W_{j,p2,d1,s3} = 1$ from 50. The transition from p_2 to p_3 is hence identified in time slot 4 from constraint 52.

Symmetry Breaking and Valid Inequalities. Inequalities 53 and 54 are added to the formulation to enforce that transfer operations between distillation units and tanks as well as shipping operations be allocated to the earlier slots of a day. As a consequence, these constraints avoid exploration of nodes of the search tree that lead to symmetric solutions.

$$\sum_{i \in I_u} X_{u,i,d,s} \geq \sum_{i \in I_u} X_{u,i,d,s+1} \quad \forall u, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (53)$$

$$\sum_p Z_{j,p,d,s} \geq \sum_p Z_{j,p,d,s+1} \quad \forall j, d, s \in \mathbf{SD}_d - \mathbf{LS}_d \quad (54)$$

For example, consider the case in which two products (p_1 and p_2) are to be shipped within the same day and there are three available time slots. The two operations would require two time slots; hence one time slot would be unallocated. Assuming that the lowest cost sequence is p_1 preceding p_2 , three symmetric feasible solutions would result in the same objective function, as shown in Table 1. By adding constraints 53 and 54, solutions 2 and 3 would become infeasible.

Table 1. Impact of Symmetry Breaking Constraints

solution	time slot 1	time slot 2	time slot 3
1	p_1	p_2	
2		p_1	p_2
3	p_1		p_2

The $\mathbf{SD}_d - \mathbf{LS}_d$ domain prevents constraints 53 and 54 to be generated for the last time slot of each day; otherwise they would include the last time slot of 1 day and the first time slot of the following one.

Valid inequalities 55 and 56 are also included to set tighter upper bounds on the summations of $VTP_{i,j,d,s}$ and $VPP_{j,p,d,s}$ respectively. These constraints are laid out in the same fashion as the tightening constraints illustrated in the work of Harjunkoski et al.²¹ for unit-specific continuous time models,

in which it is ensured that the total duration of tasks should be lower than the time horizon. The major difference is that 55 and 56 are expressed in terms of volumes and capacities, instead of task duration and time horizon.

$$\sum_{j \in J_i} \sum_{s \in \mathbf{SD}_d} VTP_{i,j,d,s} \leq HDFT_i^{\max} \quad \forall i, d \quad (55)$$

$$\sum_{p \in P} \sum_{s \in \mathbf{SD}_d} VPP_{j,p,d,s} \leq HDPF_j^{\max} \quad \forall j, d \quad (56)$$

Objective Function. The objective function is to minimize the total cost including raw material, pumping, and transition cost, which is considered the most significant cost in this work. Inventory cost was not considered because tanks simply hold intermediate products.

$$\text{cost} = \sum_i \sum_{j \in J_i} \sum_d \sum_{s \in \mathbf{SD}_d} (C_{rm_i} + C_{p_i}) VTP_{i,j,d,s} + \sum_j \sum_p \sum_{p' \neq p} \sum_{s > 1} C_{t_{j,p,p'}} TR_{j,p,p',s} \quad (57)$$

Two models are defined based on the presented formulation, namely:

M1

$$\begin{cases} \text{objective function: (57)} \\ \text{subject to} \\ \text{constraints (1–48) + (50–56)} \end{cases}$$

M2

$$\begin{cases} \text{objective function: (57)} \\ \text{subject to} \\ \text{constraints (1–48) + (50–52) + 55 + 56} \end{cases}$$

5. COMPUTATIONAL RESULTS

The formulation presented in the previous section was applied to a real-world problem that includes three distillation units $\{u_1, u_2, u_3\}$, six rundown tanks ($I_{u1} = \{i_1, i_2\}$, $I_{u2} = \{i_3, i_4\}$, $I_{u3} = \{i_5, i_6\}$), and three pipelines $\{j_1, j_2, j_3\}$ —Figure 1. Due to operating constraints, if distillation unit u_1 transfers to Tank i_1 it cannot be used to blend end products. In that case, Tank i_2 would be available. Three end products are considered consisting of three diesel types that are qualified with different specifications as to sulfur content and cetane number. The input data is

Table 2. Computational Results for M1

total number of slots GAP	12 slots (1%)	16 slots (1%)	12 slots (0%)	16 slots (0%)
continuous variables	997	1321	997	1321
binary variables	936	1248	936	1248
equations	4728	6288	4728	6288
MILP solution	1411.84	1414.21	1410.97	1410.97
RMILP solution	580.62	580.62	580.62	580.62
CPU time (min)	2.07	3.25	364.03	98.21
iterations (million)	0.48	0.65	84.26	25.17
raw material cost (\$)	20.37	22.57	19.57	19.57
pumping cost (\$)	11.47	11.64	11.40	11.40
transition cost (\$)	1380.00	1380.00	1380.00	1380.00

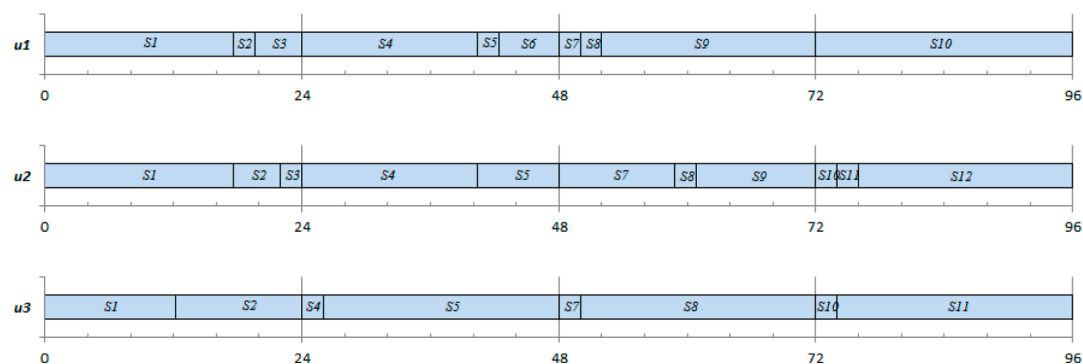


Figure 6. Gantt charts the for distillation units. Symbols inside bars refer to allocated time slots.

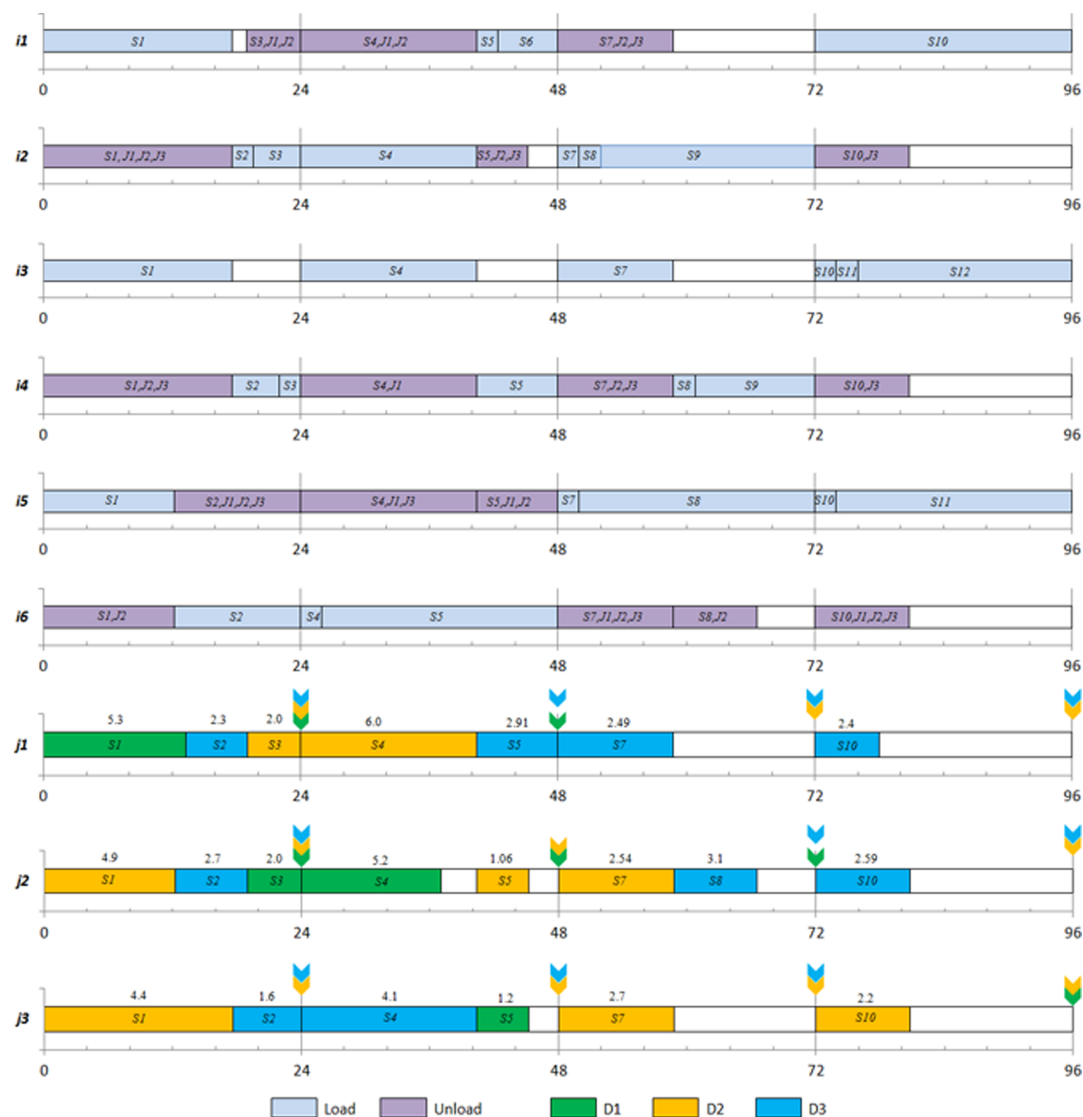


Figure 7. Gantt charts for tanks and pipelines. Symbols inside bars refer to allocated time slots and pipelines to which tanks feed. Colored arrows indicate product demand points.

presented in Appendix A. Demand is given in Table A1, and parameters related to tanks are presented in Table A2, while parameters related to distillation units and pipelines are given in Table A3. End product specs are presented in Table A4, and Table A5 presents transition costs.

Given that three products need to be blended and that in each time slot only one product can be shipped through a pipeline, 3 time slots per day were postulated resulting in a total

of 12 time slots spanning a time horizon of 4 days or 96 h, which represents the minimum number of required time slots. An instance with 16 time slots was also solved with the intent to investigate the optimal number of time slots. Each day was composed of 24 h whereas the imposed minimum operating time window was 2 h ($MH \geq 2$). The MILP problems were solved by GAMS/CPLEX 23.7 on an Intel Core i7 2.5 GHz with 8 GB of RAM. Computational results for **M1** are shown in

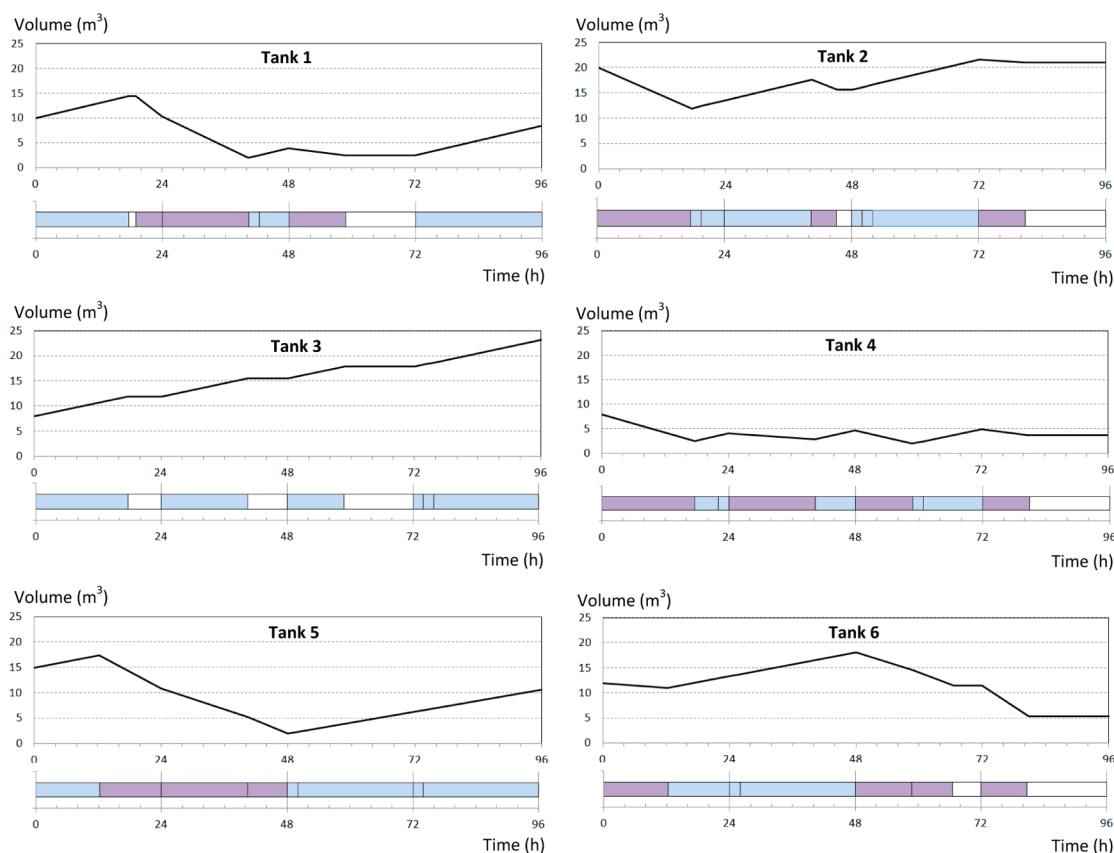


Figure 8. Tank inventory profiles.

Table 2. The results for model **M1** with the relative gap of 0% were used to determine the optimal number of time slots. Comparison of the objective function values of the two cases reveals that there is no improvement when going from 12 to 16 time slots, which indicates that 12 time slots is sufficient to guarantee optimality.¹¹ When the tolerance is relaxed to 1%, it enables us to find a nearly optimal solution (0.06% larger than the global optimal solution) in around 2 min as opposed to 6 h with 0% gap.

The Gantt charts for the distillation units are presented in Figure 6. It can be noticed that the operational rule imposing that column units should operate continuously throughout the time horizon is satisfied. Within each day, operations between columns and tanks are synchronized across the postulated time slots, which can be verified by comparing the Gantt charts of columns and associated tanks. The blue bars in Figure 7 refer to tank loading operations. Note that the blue bars of tank i_1 and tank i_2 for instance complement each other, which in turn are assigned to the same time slots of the corresponding column transfers. It can also be observed that while tank i_5 is loaded by column u_3 , tank i_6 is free to feed pipeline j_2 in time slot s_1 in the first day. Yet, because tank loading and unloading are non overlapping operations they should be assigned to different times slots, which is indeed verified for example for tank i_2 in the first day, where it feeds pipelines j_1 , j_2 , and j_3 in time slot s_1 , whereas it is loaded by column u_1 in time slot s_2 and s_3 . Another point to be observed in Figures 6 and 7 is that column and pipeline operations are always allocated to the first slots of each day, satisfying constraints S3 and S4. In order to evaluate the contribution of the symmetry breaking constraints S3 and S4, **M2** was solved considering 0% gap tolerance and 12 time slots.

No solution was reported after 6 h. The best solution reported after the 6 h was \$1940.57 with a 12.36% gap.

Further analysis of the results obtained for **M1** considering a relative gap tolerance of 0% and 12 time slots reveals that the global optimal solution was found around 22.4 million iterations, which represents 26% of the total number of iterations required to guarantee global optimality. In other words, most of the computing time is spent in refining the solution and proving global optimality.

The cost components of the objective function presented in Table 2 lead us to the conclusion that the biggest contribution to the total cost is the transition cost, which means that minimizing product interfaces is critical. Results from Figure 7 show that unnecessary product transitions in pipelines have been minimized, which means that the formulation proposed in this work for identifying transitions was effective. Even at the transition from 1 day to another, there is the common tendency of delaying product transitions as much as possible and consequently conducting to a smoother operation, which is highly desirable from an operational and costing stand point. Inventory profiles are presented in Figure 8 which also shows accordance between timing and material balance.

Day-Ahead Planning versus Optimal Planning. Consider the case in which daily schedules are produced such that the amount of each end product shipped within each day should exactly match the respective demand, here referred as day-ahead planning. Figure 9 shows the Gantt charts for the pipelines resulting from the decision making process typically followed by a scheduler based on the minimization of transition cost. It can be observed that transitions have been minimized starting the pumping operation in 1 day with the end product last pumped in the previous day. However, comparison of the

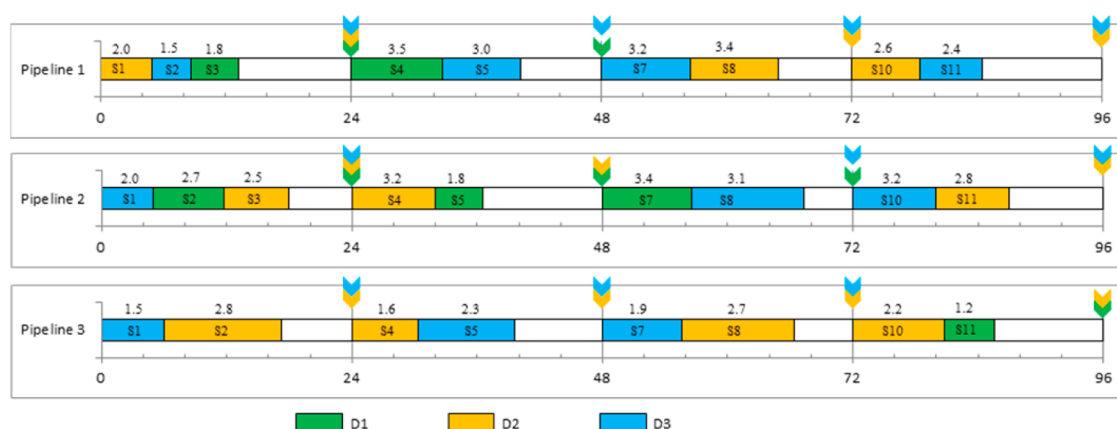


Figure 9. Pipeline Gantt charts for the one-day planning. Symbols inside bars refer to allocated time slots. Colored arrows indicate product demand points.

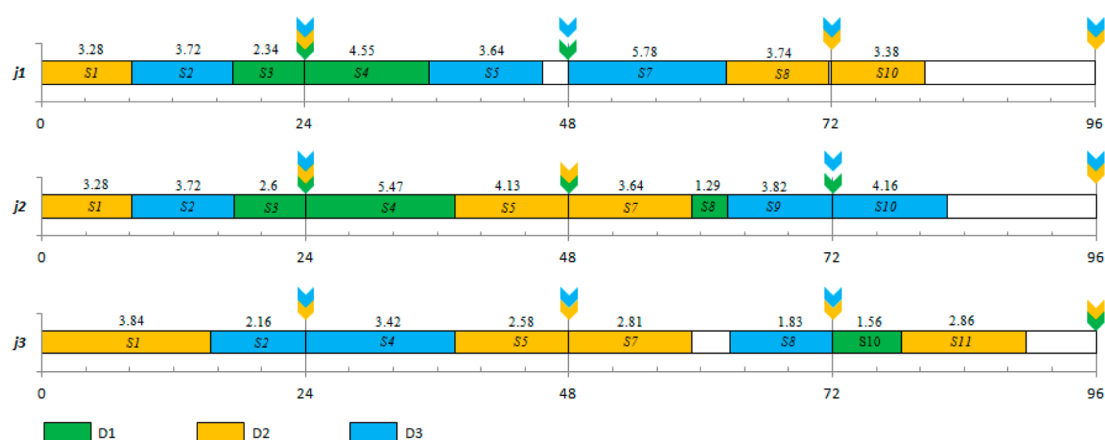


Figure 10. Pipeline Gantt charts for high demand-day-ahead planning. Symbols inside bars refer to allocated time slots. Colored arrows indicate product demand points.

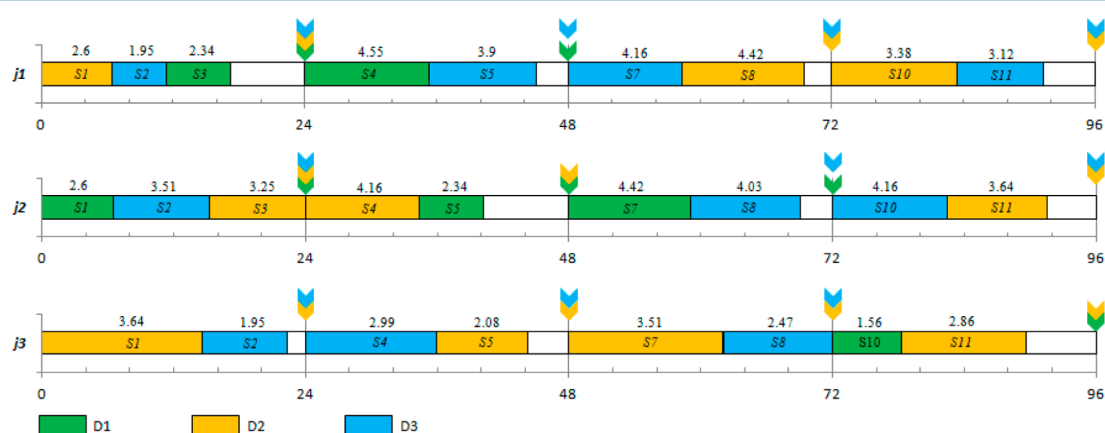


Figure 11. Pipeline Gantt charts for high demand-optimal planning. Symbols inside bars refer to allocated time slots. Colored arrows indicate product demand points.

day-ahead planning approach with the solution obtained with the optimization considering information on the whole time horizon—optimal planning—shows that because parcel volumes shipped cannot include amounts representing anticipation of future demands, there is no opportunity to reduce the number of transitions. While 14 transitions were required by the one-day planning strategy, which corresponds to \$2,090.00 total transition cost, 10 transitions were required by the optimal planning, which corresponds to \$1,380.00 total transition cost,

resulting in 34% savings. This analysis illustrates the importance of creating schedules that anticipate longer term demand.

High Demand. Consider a scenario of high demand, in which the values presented in Table A1 are increased by 30%. As can be seen by the Gantt charts in Figure 10, pipeline utilization is increased as a consequence of the high demand, whereas the free capacity is decreased in this scenario. With increased utilization there is less opportunity for the long-term planning to optimize transitions. Note in Figure 11 that 15 transitions were required by

the day-ahead planning strategy, which corresponds to \$2,190.00 total transition cost. The optimal planning, on the other hand, required 14 transitions, which corresponds to \$2,000.00 total transition cost, resulting in only 0.087% savings.

6. CONCLUSIONS

In the present work, the diesel blending and distribution scheduling problem was addressed with a hybrid time representation, which was shown suitable for meeting intermediate due dates, while taking advantage of the continuous time flexibility for accommodating transfer operations. Moreover, with the timing variables being defined over units and time slots it was possible to guarantee that the in-line blending produced homogeneous end products by forcing tanks to be aligned to pipelines for the duration of the pumping operation. Solution time was dramatically decreased by adding valid inequalities for symmetry breaking. Two scenarios were analyzed to illustrate the importance of the short-run optimized schedule as opposed to the one-day planning.

In real-world scenarios, intermediate product qualities are usually dependent on the petroleum mix charged to distillation units as well as on operating conditions. This would turn the proposed formulation in a nonlinear optimization problem, a subject that is to be explored in future developments.

■ APPENDIX A

Tables A1–A5 showing data as mentioned in the main text.

Table A1. Demand $\times 10^3$ (m³)

		day 1	day 2	day 3	day 4
pipeline 1	D1	1.8	3.5		
	D2	2.0		3.4	2.6
	D3	1.5	3.0	3.2	2.4
pipeline 2	D1	2.0	1.8	3.4	
	D2	2.5	3.2		2.8
	D3	2.7		3.1	3.2
pipeline 3	D1				1.2
	D2	2.8	1.6	2.7	2.2
	D3	1.5	2.3	1.9	

Table A2. Data Related to Tanks

	tank 1	tank 2	tank 3	tank 4	tank 5	tank 6
sulfur (wt %)	0.30	0.30	0.60	0.40	1.00	1.00
cetane number	42.0	42.0	40.3	39.0	40.0	40.0
$V_i^{\min} \times 10^3$ (m ³)	2	2	2	2	2	2
$V_i^{\max} \times 10^3$ (m ³)	30	30	30	30	30	30
$V_i^0 \times 10^3$ (m ³)	10	20	8	8	15	12
$FT_i^{\min}$ (m ³ /h)	30	30	40	40	40	40
$FT_i^{\max}$ (m ³ /h)	500	500	500	500	500	500
Cp_i (\$/m ³)	0.20	0.20	0.18	0.18	0.16	0.16
Crm_i (\$/m ³)	0.60	0.60	0.40	0.40	0.05	0.05

Table A3. Data Related to Columns and Pipelines

	column 1	column 2	column3
$FC_u^{\min}$ (m ³ /h)	250	220	180
$FC_u^{\max}$ (m ³ /h)	300	250	200
	pipeline 1	pipeline 2	pipeline 3
$FP_j^{\min}$ (m ³ /h)	50	50	50
$FP_j^{\max}$ (m ³ /h)	400	400	250

Table A4. Diesel Specs

diesel grade	sulfur (wt %)	cetane number
D1	0.3	42
D2	0.5	40
D3	0.1	40

Table A5. Transition Costs (\$)

	D1	D2	D3
D1		110	100
D2	130		120
D3	190	190	

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Notes

The authors declare no competing financial interest.

■ NOMENCLATURE

Indices

d = days
 i = tanks
 j = pipelines
 k = product specs
 p = end products
 s = time slots
 u = distillation units

Sets

D = days
 DS_s = day to which slot s belong to
 FS_d = first slot of day d
 I = tanks
 I_j = tanks that can feed pipeline j
 I_u = rundown tanks i to which column u can discharge
 J = pipelines
 J_i = pipelines to which tank i can discharge
 K = product specs
 LS_d = last slot of day d
 P = end products
 S = time slots
 SD_d = slots that belong to day d
 U = distillation units
 U_i = columns that can feed tank i

Parameters

Crm_i = cost of intermediate product consumed from tank i
 Cp_i = pumping cost from tank i
 $Ct_{j,p,p'}$ = cost incurred due to an interface created between product p and p' in pipeline j
 $DI_{j,p,d}$ = One if there is demand incidence for product p at the end of day d at pipeline j
 $D_{j,p,d}$ = demand for product p at the end of day d at pipeline j
 $FC_u^{\max}$ = maximum production of distillation unit u
 $FC_u^{\min}$ = minimum production of distillation unit u
 $FP_j^{\max}$ = upper bound on the volume flow rate for pipeline j
 $FP_j^{\min}$ = lower bound on the volume flow rate for pipeline j
 $FT_i^{\max}$ = upper bound on the volume flow rate from tank i
 $FT_i^{\min}$ = lower bound on volume flow rate from tank i
 H = time horizon duration in hours
 HD = number of hours in a day
 K_{ik} = quality k maintained in tank i

MH = minimum number of hours assigned to an operation
 NC_i = maximum number of simultaneous connections that tank i can have to pipelines
 $SPC_{p,k}$ = spec value of quality k for product p
 T_d^f = first hour corresponding to day d
 T_d^l = last hour corresponding to day d
 VD_i^0 = initial storage volume in tank i
 $V_i^{\max}$ = storage capacity of tank i
 $V_i^{\min}$ = lower bound on the volume stored in tank i
 V_u^{up} = upper bound on the total volume produced by column u in 1 day
 V_j^{up} = upper bound on the total volume shipped through pipeline j in 1 day
 VT_i^{up} = upper bound on the total volume transferred from tank i in 1 day

Continuous Variables

$TC_{u,d,s}^f$ = end time at which distillation unit u releases intermediate product to rundown tanks in time slot s of day d
 $TC_{u,d,s}^s$ = start time at which distillation unit u releases intermediate product to rundown tanks in time slot s of day d
 $TP_{j,d,s}^f$ = end time at which a product is pumped through pipeline j in time slot s of day d
 $TP_{j,d,s}^s$ = start time at which a product is pumped through pipeline j in time slot s of day d
 $TT_{i,d,s}^f$ = end time at which tank i execute an operation (load or unload) in time slot s of day d
 $TT_{i,d,s}^s$ = start time at which tank i execute an operation (load or unload) in time slot s of day d
 $VCT_{u,i,d,s}$ = volume transferred between column u and tank i in time slot s
 $VD_{i,d}$ = volume of material contained in tank i at the end of day d
 $VPP_{j,p,d,s}$ = volume of product p transported through pipeline j in time slot s
 $VS_{i,d,s}$ = volume of material contained in tank i at the end of time slot s of day d
 $VTP_{i,j,d,s}$ = volume transferred between tank i and pipeline j in time slot s
 $TR_{j,p,p',s}$ = One if product p' is pumped right after product p in pipeline j in time slot s and 0 otherwise

Binary Variables

$X_{u,i,d,s}$ = 1 if column u feeds tank i in time slot s of day d
 $Y_{i,j,d,s}$ = 1 if tank i is used to compose the product shipped through pipeline j in time slot s of day d
 $Y_{i,j,d,s}^f$ = 1 if tank i feeds pipeline j in time slot s of day d and the end time of pipeline j correspond to the end time of tank i
 $Y_{i,j,d,s}^s$ = 1 if tank i feeds pipeline j in time slot s of day d and the start time of pipeline j correspond to the start time of tank i
 $Z_{j,p,d,s}$ = 1 if product p is blended and shipped through pipeline j in time slot s of day d
 $W_{j,p,d,s}$ = 1 if no product is blended and shipped through pipeline j in time slot s of day d , but the last product shipped was product p

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